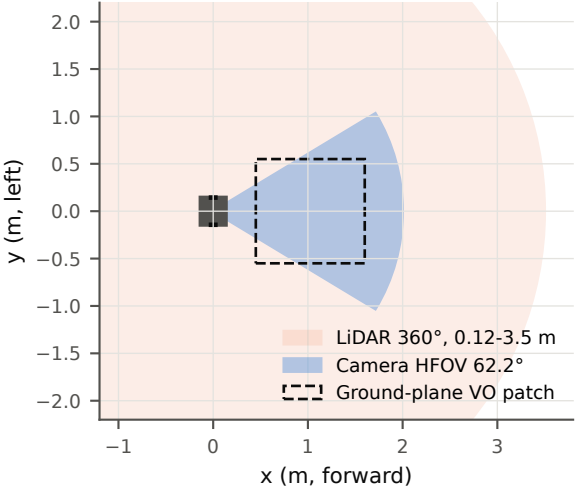
(a) Sensor footprint (top view)



(b) Simulated sensor suite (design parameters)

Sensor	Rate	Model (Table II)
IMU	100 Hz	gyro σ 0.003 rad/s,
Wheel enc.	50 Hz	4096 ticks/rev
Camera	15 Hz	160×96 px, h=0.12
LiDAR	5 Hz	360 beams, σ 0.01
Base	—	r=0.033 m, b=0.28